

Ch. 6

Ch. 5 rev:

- can use scattering to determine particles that consist of how they interact

Note:

- used colliders in time $\rightarrow$ $L \neq L^\dagger$ but can do asymptotic states w/ $L \rightarrow \infty$ $L^\dagger \rightarrow \infty$

LSZ S-matrix reduction formula:

$$\left(\langle f | S | i \rangle = \left[i \int d^4x_1 e^{-i p_1 x_1} (\square + m^2) \right] \dots \left[i \int d^4x_n e^{i p_n x_n} (\square + m^2) \right] \right. \\ \left. \times \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(x_3) \dots \phi(x_n) \} | \Omega \rangle \right)$$

i is exp. for in, i for final

Note:

function of $\square + m^2$ "project onto the S-matrix"

$$(\square + m^2 \rightarrow -p^2 + m^2)$$

$\therefore$ These terms delete terms from higher-order pert. except w/ poles at $\frac{1}{p^2 - m^2}$

$$\left(\frac{1}{p^2 - m^2} \right)$$

$\sim 2, 1, 0$

con. to on shell particles

$\rightarrow$ have fixed S-matrix projects "one-particle asymptotic states from higher-order pert. of fields"

place $(2 \rightarrow n)$ arguments a_i (class)

$$d\sigma = \frac{1}{(2E_1)(2E_2)|\vec{v}_1 - \vec{v}_2|} |M|^2 dT_{1,2,3,4}$$

s- matrix:

$$\langle f | S | i \rangle = i(2\pi)^4 \delta^4(p_f) M$$

in/i at $-\infty$, & out ∞

i.e.

$$|i\rangle = \sqrt{2\pi\omega_1} \sqrt{2\pi\omega_2} a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) |\Omega\rangle$$

$$|f\rangle = \sqrt{2\pi\omega_3} \dots \sqrt{2\pi\omega_n} a_{p_3}^\dagger(\infty) \dots a_{p_n}^\dagger(\infty) |\Omega\rangle$$

$|f\rangle \neq |i\rangle$ b.c. scattering actually happens,

$$u(\omega) = \sqrt{\frac{2\pi}{\omega}} e^{i\omega t}$$

$|\Omega\rangle$ ground state

q matrix:

$$\langle P | S | i \rangle = i \sqrt{\omega_1 \omega_2 \dots \omega_n}$$

$$\times \langle \Omega | a_{p_3}(\infty) \dots a_{p_n}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) |\Omega\rangle$$

- not very useful

- must use 4th quantum fields

$$\phi(x) = \phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \left[a_p e^{ipx} + a_p^\dagger e^{-ipx} \right]$$

Heisenberg picture.

- use operator to create states at same time

$i\hbar$ time independent,

$$a_p(t) = e^{iH(t-t_0)} a_p(t_0) e^{-iH(t-t_0)}$$

$$\phi(\vec{p}, t) = e^{iH(t-t_0)} \phi(\vec{p}, t_0) e^{-iH(t-t_0)}$$

All need to know about ϕ & a_p are they commute at any any times

$$\langle \Omega | \phi(\vec{p}, t = \pm\infty) | \sigma \rangle = \langle e^{i\vec{p} \cdot \vec{r}} \rangle \text{ w/ mod. } (-)$$

$$i \int d^4x e^{i\vec{p} \cdot \vec{x}} (\Box + m^2) \phi(x) = \sqrt{2\pi p} [a_p(\infty) - a_p(-\infty)]$$

$$m | \vec{p}^m = (\omega_p, \vec{p})$$

proof:

$$\begin{aligned} i \int d^4x e^{i\vec{p} \cdot \vec{x}} (\Box + m^2) \phi(x) &= i \int d^4x e^{i\vec{p} \cdot \vec{x}} (\partial_\mu^2 - \vec{\partial}_x^2 + m^2) \phi \\ &= i \int d^4x e^{i\vec{p} \cdot \vec{x}} (\partial_\mu^2 + \vec{p}^2 + m^2) \phi(x) \\ &= i \int d^4x e^{i\vec{p} \cdot \vec{x}} (\partial_\mu^2 + \omega_p^2) \phi(x) \end{aligned}$$

$$\begin{aligned} \phi \quad \int d^4x e^{i\vec{p} \cdot \vec{x}} (i\partial_\mu + \omega_p) \phi(x) &= [i\omega_p e^{i\vec{p} \cdot \vec{x}} (i\partial_\mu + \omega_p) + e^{i\vec{p} \cdot \vec{x}} (i\partial_\mu + \omega_p)] \phi(x) \\ &= i e^{i\vec{p} \cdot \vec{x}} (\partial_\mu^2 + \omega_p^2) \phi(x) \end{aligned}$$

$$i \int d^4x e^{ipx} (D+m) \phi(x) = \int d^4x \partial_t \left[e^{ipx} (i\partial_t + m) \phi(x) \right]$$

$$= \int d^4x \partial_t \left[e^{ipx} \int d^3z e^{-i\vec{p} \cdot \vec{z}} (i\partial_t + m) \phi(x) \right]$$

- True for all ϕ (bounded function w/out dirac)

- Total time derivative so only depends on $t = \pm \infty$

cause of ϕ of quantum field.

$$\int d^3x e^{-i\vec{p} \cdot \vec{z}} (i\partial_t + m) \phi(x)$$

$$= \int d^3x e^{-i\vec{p} \cdot \vec{z}} (i\partial_t + m)$$

$$\times \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k(t) e^{-ikx} + a_k^\dagger(t) e^{ikx} \right)$$

???

$$= \int \frac{d^3k}{(2\pi)^3} \int d^3z \left[\left(\frac{\omega_k + m}{\sqrt{2\omega_k}} \right) a_k(t) e^{-ikx} e^{-i\vec{p} \cdot \vec{z}} + \left(\frac{-\omega_k + m}{\sqrt{2\omega_k}} \right) a_k^\dagger(t) e^{ikx} e^{-i\vec{p} \cdot \vec{z}} \right]$$

Math Note:

used $\partial_t a_k(t) = 0$ at $t = \pm \infty$

Note: $\int d^3x$ integral enforces $\omega_k = \omega_p$ via δ -function

$$\int d^3x e^{-i\vec{p} \cdot \vec{z}} (i\partial_t + m) \phi(x) = \sqrt{2\omega_p} a_p(t) e^{-i\omega_p t}$$

$$\rightarrow i \int d^4x e^{i\phi(x)} (\Box + m^2) \phi(x) = \int dt \partial_t \left[(e^{i\phi(x,t)}) (\sqrt{-g} g^{00} \phi(x,t) e^{i\phi(x,t)}) \right]$$

$$= \sqrt{-g} g^{00} [a_p(\infty) - a_p(-\infty)]$$

h.c. implies

$$\sqrt{-g} g^{00} [a_p^\dagger(\infty) - a_p^\dagger(-\infty)] = -i \int d^4x e^{-i\phi(x)} (\Box + m^2) \phi(x)$$

we can compute $\langle p|q|i \rangle = \sqrt{a_1 a_2 \dots a_n}$

$$\times \langle \Omega | a_{p_1}(\infty) \dots a_{p_n}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle$$

Note $a_{p_1}(\infty) \dots a_{p_n}(\infty) \times a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty)$ already fixed - ordered

since all fin. states have a_p at ∞ , & initial states have a_p at ∞

time ordering lets ^{???}

$$\langle p|s|i \rangle = \sqrt{a_1 \dots a_n} \langle \Omega | T \left\{ [a_{p_1}(\infty) - a_{p_1}(-\infty)] \dots [a_{p_n}(\infty) - a_{p_n}(-\infty)] \right.$$

$$\left. \times [a_{p_1}^\dagger(-\infty) - a_{p_1}^\dagger(\infty)] [a_{p_2}^\dagger(-\infty) - a_{p_2}^\dagger(\infty)] \right\} | \Omega \rangle$$

"T acts on arguments $a_p(x)$ & left & annihilate w/ $\langle p|$ & $a_p(-\infty)$ operators which annihilate $a_p|i \rangle$ "

$$\rightarrow \langle p_3 \dots p_n | s | p_1 p_2 \rangle = \left[i \int d^4x_1 e^{-i\phi(x_1)} (\Box_1 + m^2) \right] \dots \left[i \int d^4x_n e^{i\phi(x_n)} (\Box_n + m^2) \right]$$

$$\times \langle \Omega | T \{ \phi(x_1) \phi(x_2) \dots \phi(x_n) | \Omega \rangle$$

which is LST reduction
(note: $\square_i^m = (\frac{\partial}{\partial x_i^m})^2$)

Notes / Discussion:

- mult. dimensional prod. of fields w/ $\square_{\mu\nu}^2$ & fermion

- FF & free, $(\square_{\mu\nu})\phi = 0$

- will be factors of propagator $\frac{1}{\square_{\mu\nu}}$

- LST "guarantees zeroes & zeroes cancel")

- LST kills terms w/o divergence

LST: "isolates asymptotic states by adding a suitably constructed zero to cancel anything that does not correspond to the state we want."

- LST nontrivial

LST for operators:

- LST powered
→ applies to operators

LGZ implies ???

$$\langle p_3 \dots p_n | g | p_1 p_2 \rangle = \left[i \int d^4 x_1 e^{-i p_1 x_1} (\square_1 + m^2) \right] \\ \times \dots \times \left[i \int d^4 x_n e^{i p_n x_n} (\square_n + m^2) \right] \\ \times \langle \Omega | \phi_1(x_1) \phi_2(x_2) \dots \phi_n(x_n) | \Omega \rangle$$

w/ $\phi_i(x)$ operators that create one particle states

$$\langle p | \phi(x) | \Omega \rangle = \sum e^{i p x}$$

- LGZ does not distinguish elementary particles

- particles w/ fields in $\mathbb{R}^4$

- "anything that overlaps with one-particle states
will produce pdf accessible by DTM=11"

(pg 74) ???

operators ϕ

$$\text{basis states } |\psi_n\rangle \text{ w/ } C_{nm} = \langle \psi_n | \phi | \psi_m \rangle$$

$$\rightarrow \phi = \sum_{n,m} \int d q_1 \dots d q_n d p_1 \dots d p_m a_1^\dagger \dots a_n^\dagger a_m \dots a_1 \\ \times C_{nm}(q_1, \dots, p_m)$$

- most important operators
 Lor. inv. composite constructed from elem. fields
 such as $O(a) = \phi(x) \partial_m \phi$

vs $z \rightarrow \infty$

- positronium bound state

positronium = electron, positron, binding photons
 & same quantum HS as $O_p = \bar{\psi}(x) \psi(x)$
 , so $\psi_f(x)$ nonzero overlap positronium

concept: ~ There don't have to be fundamental fields
 associated w/ every asymptotic state to
 calculate S-matrix

- If don't know elementary particles, can introduce terms
 in Lagrangian for S-matrix

- Technique referred to as effective field
 theory

Free Field

- have method for quantization
 - Just add time ordered products
- Free theory:

start w/ free field:

$$\phi_0(x, t) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (a_k e^{-ikx} + a_k^\dagger e^{ikx})$$

w/ $\omega_k = \omega_k = \sqrt{m^2 + k^2}$ & a_k & $a_k^\dagger$ time independent

Then

$$\langle 0 | \phi_0(x_1) \phi_0(x_2) | 0 \rangle = \int \frac{d^3k_1}{(2\pi)^3} \int \frac{d^3k_2}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{k_1}}} \frac{1}{\sqrt{2\omega_{k_2}}} \langle 0 | a_{k_1} a_{k_2}^\dagger | 0 \rangle e^{i(k_2 x_2 - k_1 x_1)}$$

w/ $|0\rangle$ ground state vacuum

since $\langle 0 | a_{k_1} a_{k_2}^\dagger | 0 \rangle = (2\pi)^3 \delta^3(\vec{k}_1, -\vec{k}_2)$

$$\langle 0 | \phi_0(x_1) \phi_0(x_2) | 0 \rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{ik(x_2 - x_1)}$$

with $\langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) \} | 0 \rangle = \langle 0 | \phi_0(x_1) \phi_0(x_2) | 0 \rangle \theta(t_1 - t_2)$

$$= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \left[e^{ik(x_2 - x_1)} \theta(t_1 - t_2) + e^{ik(x_1 - x_2)} \theta(t_2 - t_1) \right]$$

$$= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \left[e^{i\vec{k}(\vec{x}_1 - \vec{x}_2)} e^{-i\omega_k \tau} \theta(\tau) + e^{-i\vec{k}(\vec{x}_1 - \vec{x}_2)} e^{i\omega_k \tau} \theta(-\tau) \right]$$

$$w/ \tau = t_0 - t_n$$

$$n \rightarrow -k \text{ fermion } ? ?$$

$$\text{times } \int d^3k, \text{ over}$$

$$\rightarrow \langle 0 | T \{ \phi_0(x) \phi_0(x') \} | 0 \rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{-i\vec{k} \cdot (\vec{x}_1 - \vec{x}_2)} \\ \times [e^{i\omega_k \tau} \theta(-\tau) + e^{-i\omega_k \tau} \theta(\tau)]$$

- The terms of sum are advanced & retarded propagators from before

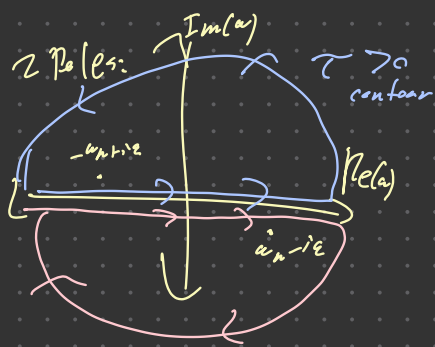
Note: Math tool kit:

$$e^{i\omega_k \tau} \theta(\tau) + e^{-i\omega_k \tau} \theta(-\tau) = \lim_{\epsilon \rightarrow 0} \frac{-2\omega_k}{2\pi i} \\ \times \int_{-\infty}^{\infty} \frac{du}{\omega_k^2 - u^2 + i\epsilon} e^{i\omega_k \tau}$$

Derivation of identity (Partial Fractions)

$$\frac{1}{\omega^2 - \omega_k^2 + i\epsilon} = \frac{1}{[\omega - (\omega_k - i\epsilon)] [\omega - (-\omega_k + i\epsilon)]} \\ = \frac{1}{2\omega_k} \left[\frac{1}{\omega - (\omega_k - i\epsilon)} - \frac{1}{\omega - (-\omega_k + i\epsilon)} \right]$$

(dropped from $\theta(\tau)$) & note $2\omega_k = 1$
since $\epsilon \rightarrow 0$)



poles at $\pm a_n \mp i\epsilon$

$\tau > 0$, close upward
 $\tau < 0$, close downward
 for the
 integration
 but
 always

cont. add $e^{ia\tau}$ on RHS

Integrate from $a \in \mathbb{R}$

$$\int_{-\infty}^{\infty} \frac{da}{a - (a_n - i\epsilon)} e^{ia\tau} = -2\pi i e^{-a_n \tau} \theta(-\tau) + O(\epsilon)$$

(Mixing sign from contour int. dir)

$$\int_{-\infty}^{\infty} \frac{da}{a - (-a_n + i\epsilon)} e^{ia\tau} = 2\pi i e^{-a_n \tau} \theta(\tau) + O(\epsilon)$$

then
 $\lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{da}{a^2 - a_n^2 + i\epsilon} e^{ia\tau} = -\frac{2\pi i}{2a_n} \left[e^{i a_n \tau} \theta(-\tau) + e^{-i a_n \tau} \theta(\tau) \right]$

$$\rightarrow \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) \} | 0 \rangle = \lim_{\epsilon \rightarrow 0} \int \frac{d^4 k}{(2\pi)^4} \frac{1}{2\omega_k} e^{-i\tilde{k}(\vec{x}_1 - \vec{x}_2)} \\ \times \int d\omega \frac{-2\omega_k}{2\pi i} \frac{1}{\omega^2 - \omega_k^2 + i\epsilon} e^{i\omega\tau}$$

The Feynman propagator is then

$$D_F = \langle 0 | T \{ \phi_0(x_1) \phi_0(x_2) \} | 0 \rangle = \left(\lim_{\epsilon \rightarrow 0} \right) \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 - m^2 + i\epsilon} e^{ik(x_1 - x_2)}$$

- Pole at $k^2 = m^2$
- Pole at $\sqrt{k^2 + m^2}$ or can be off-shell
- i from contour integral
- ϵ "just added for representing same ordering in a simple way"
- $\epsilon \rightarrow 0$ in end
- $i\epsilon$ always works for time-ordered prod.
- $\epsilon=0$ Feynman p.p. is the Green function for k.g. eqn

$$(\square + m^2) D_F(x, y) = -i \delta^4(x, y)$$

- It's
- "just computing classical properties in really complicated way"